Glycoscience Analaysis, Part 1

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Proteomics and Biomarkers_BIOC_6237

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Today’s Class

- Review glycobiology basics
- Introduce a specific use case
- Walk through proteomics analysis

Glycobiology Overview: Cells

- Study of glycans
  - Sugars attached to proteins or cells
  - The specific composition of the sugars can have major effects on that sugar or cell
- Was originally thought to be a protective coating
  - “Glycocalyx” terminology
  - Now thought to be more like a language
- Every organism covers its cells with glycans
  - Glycans mediate cell-cell interactions (e.g. immune cells)
- Involved in nearly every biological process in the body
  - Can take many different forms

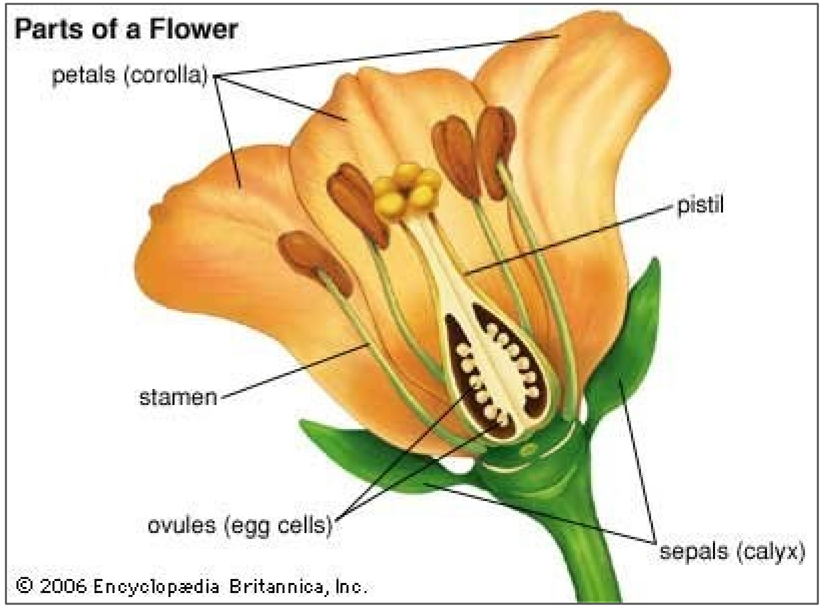

Image credit: Encyclopedia Brittanica

Glycobiology Overview: Protein Glycosylation

- The majority of proteins are glycosylated
  - Estimated that >50% of mammalian proteins are appended with a glycan[1]
  - Nearly all secreted proteins
    - Further supports "communication" role for glycans
  - Mammalian cells may have as many as 10 million glycans attached to proteins[2]
  - Some proteins highly glycosylated (>50%)
    - E.g. nucleoporins

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  - Some proteins highly glycosylated (>50%)
    - E.g. nucleoporins
- Immune functions are largely mediated through protein-glycan interactions, which has implications for:
  - Physiology
  - Pathogen recognition
  - Cancer
  - Autoimmune diseases

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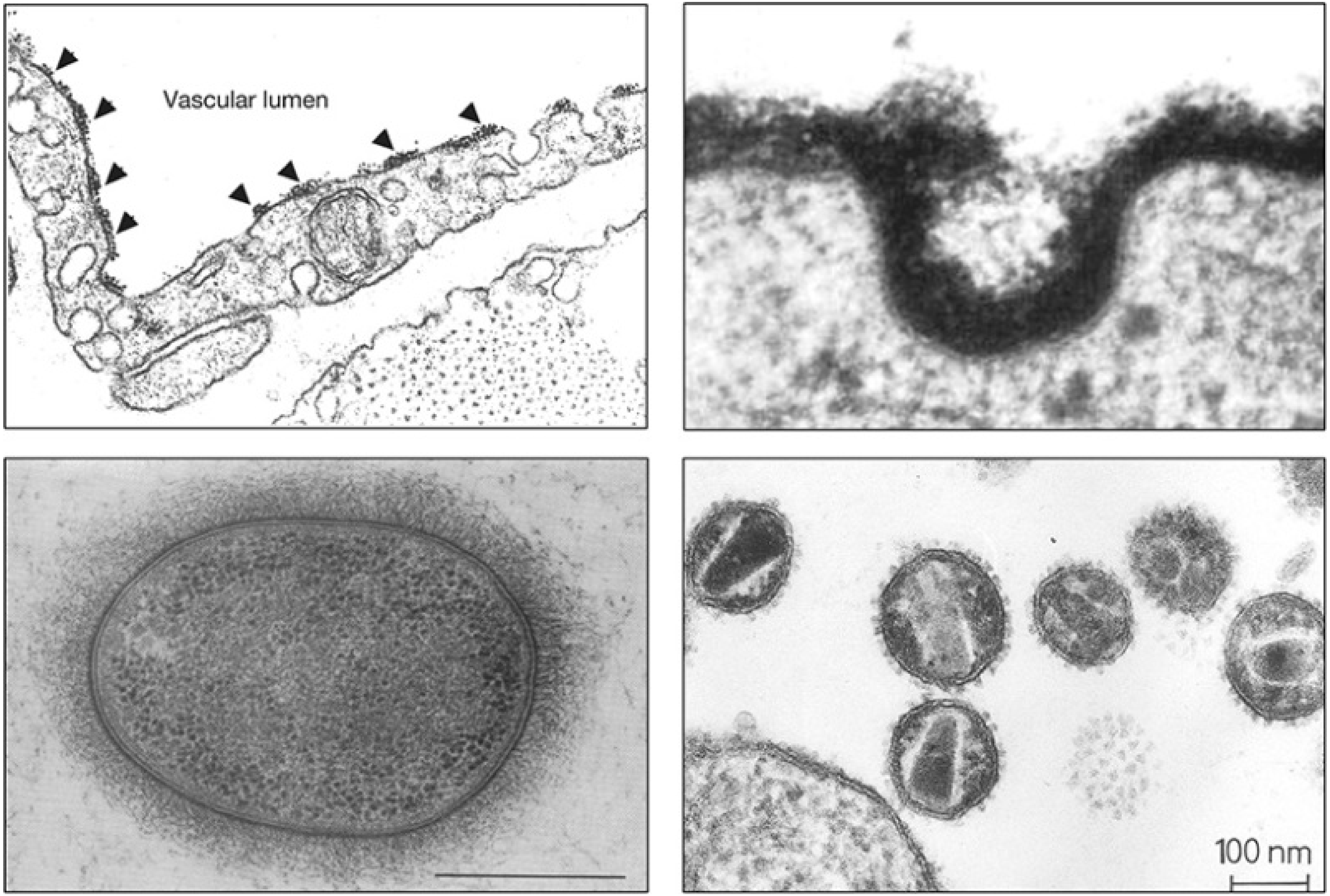

Ajit Varki and Stuart Kornfeld. Essentials of Glycobiology, 3rd edition. Cold Spring Harbor Press; 2015-2017 [Internet]. Figure 1.4

Glycoconjugate Functions: Examples

- Identity marker
  - Interface for transfusions

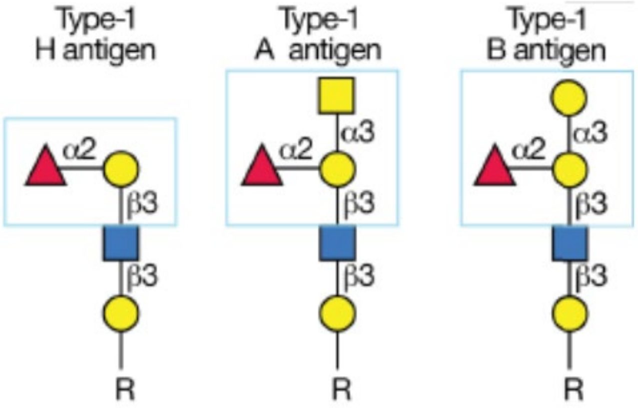

Image: Varki wet al. 2009. Essentials of Glycobiology

Glycoconjugate Functions: Examples

- Identity marker
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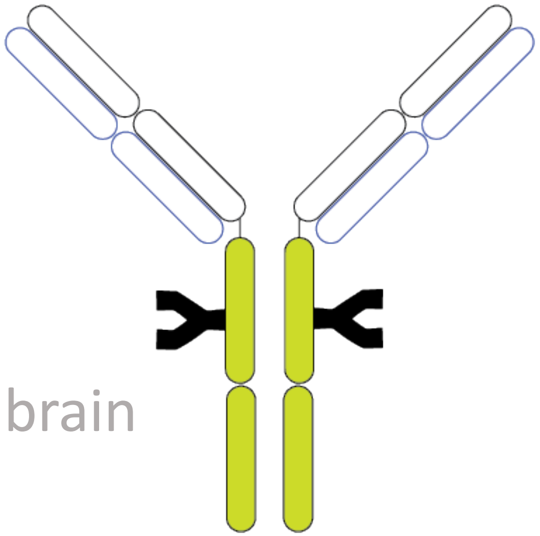
- Important for blood flow[1]
  - May contribute to the way that blood flow is controlled in the brain

1: J Cereb Blood Flow Metab. 2000 Nov;20(11):1571-8

Glycoconjugate Functions: Examples



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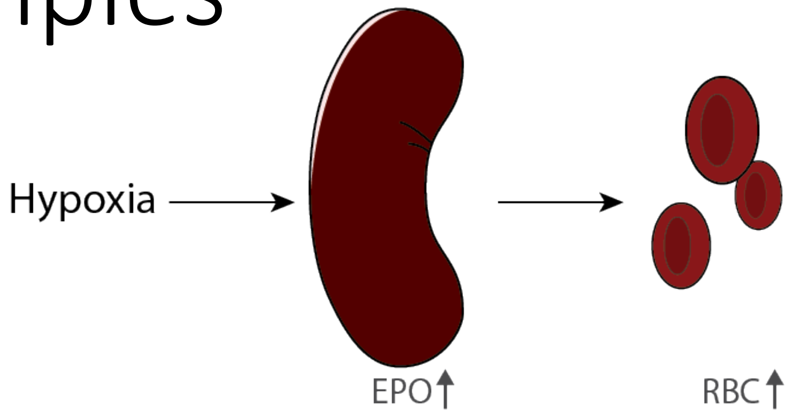

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- Antibodies
  - Fab region binds a target (e.g. cancer epitope), then Fc region binds a second receptor to stimulate biological event
- Next generation of therapeutic and diagnostic targets
  - Most new biological drugs are glycosylated proteins
  - Erythropoietin
  - Must be properly glycosylated!

Pauline Rudd

Glycans in Disease

- Involved in many diseases
  - Infectious disease
    - Many pathogenic bacteria
    - Almost all viruses interact via the carbohydrates on the surface
  - Cancer
    - All cancers express different carbohydrates on their surfaces compared to their normal tissue counterparts, and glycosylation of their secreted proteins is also different

Glycans are dynamic… and mysterious

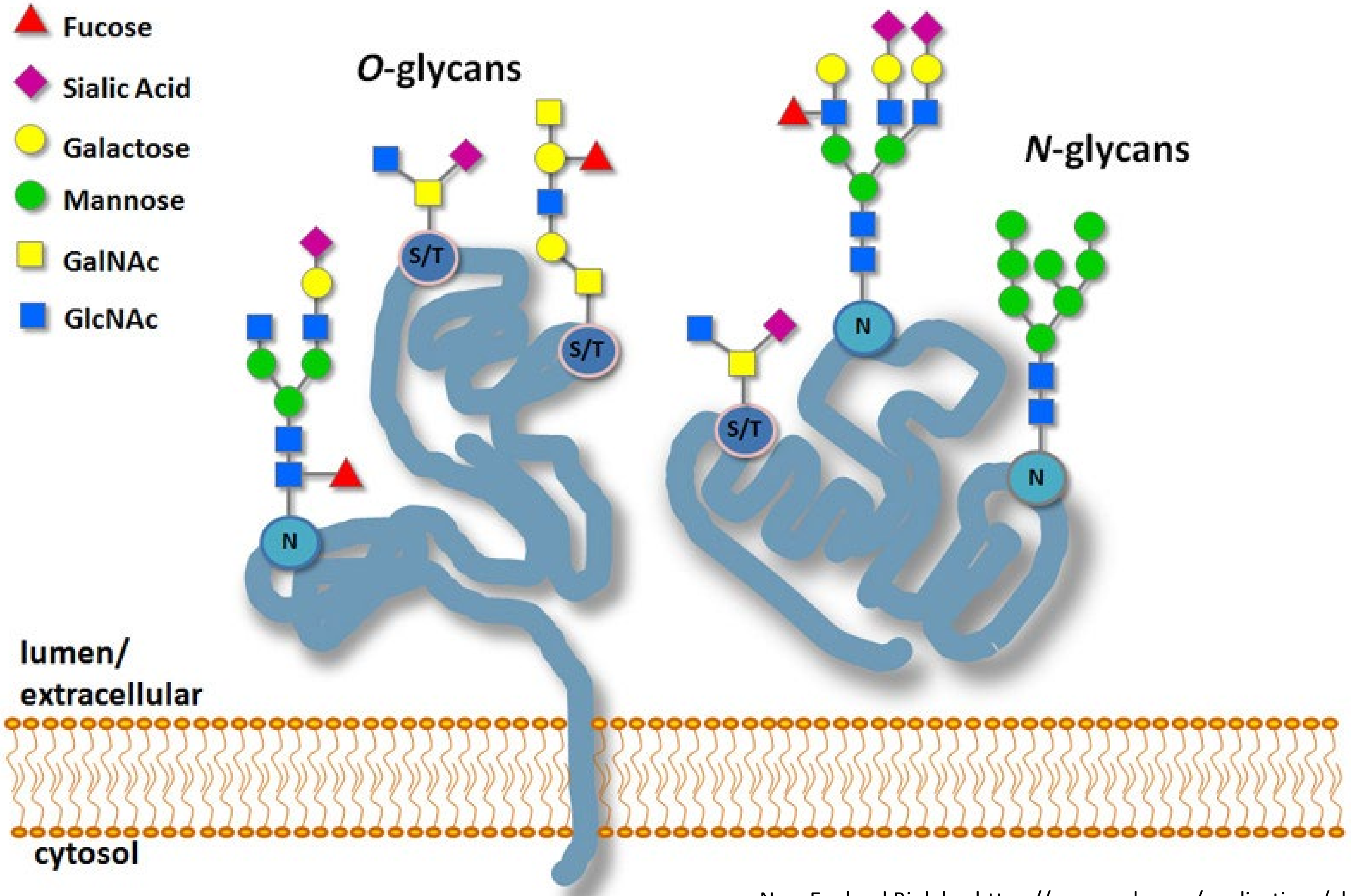
- Glycan structures are diverse and specific to cell type
- Glycan structures are dynamic
  - Glycans change with healthy aging and disease
  - Can change (rapidly) in response to a change in the environment
    - Can induce glycosylation changes by changing cell culture conditions
- Despite being omnipresent and involved in so many functions, little is known about them!
  - Big target for future therapeutics

Glycans as Biomarkers

- "In many diseases that we studied, including diabetes, cancer, and IVT, changes in the composition in either total plasma glycan or IgG glycan contained more information about the disease than all other clinical parameters combined. Therefore, glycan profiling is definitely a very promising biomarker."

  - Gordan Lauc, Ph.D. University of Zagreb

Flavors of Glycans: N and O



New England Biolabs: https://www.neb.com/applications/glycobiology-and-proteomics/glycobiology

Glycosylation Sites

N-linked glycosylation site:
PNGase F recognition site:

NXS/T

X = anything but proline

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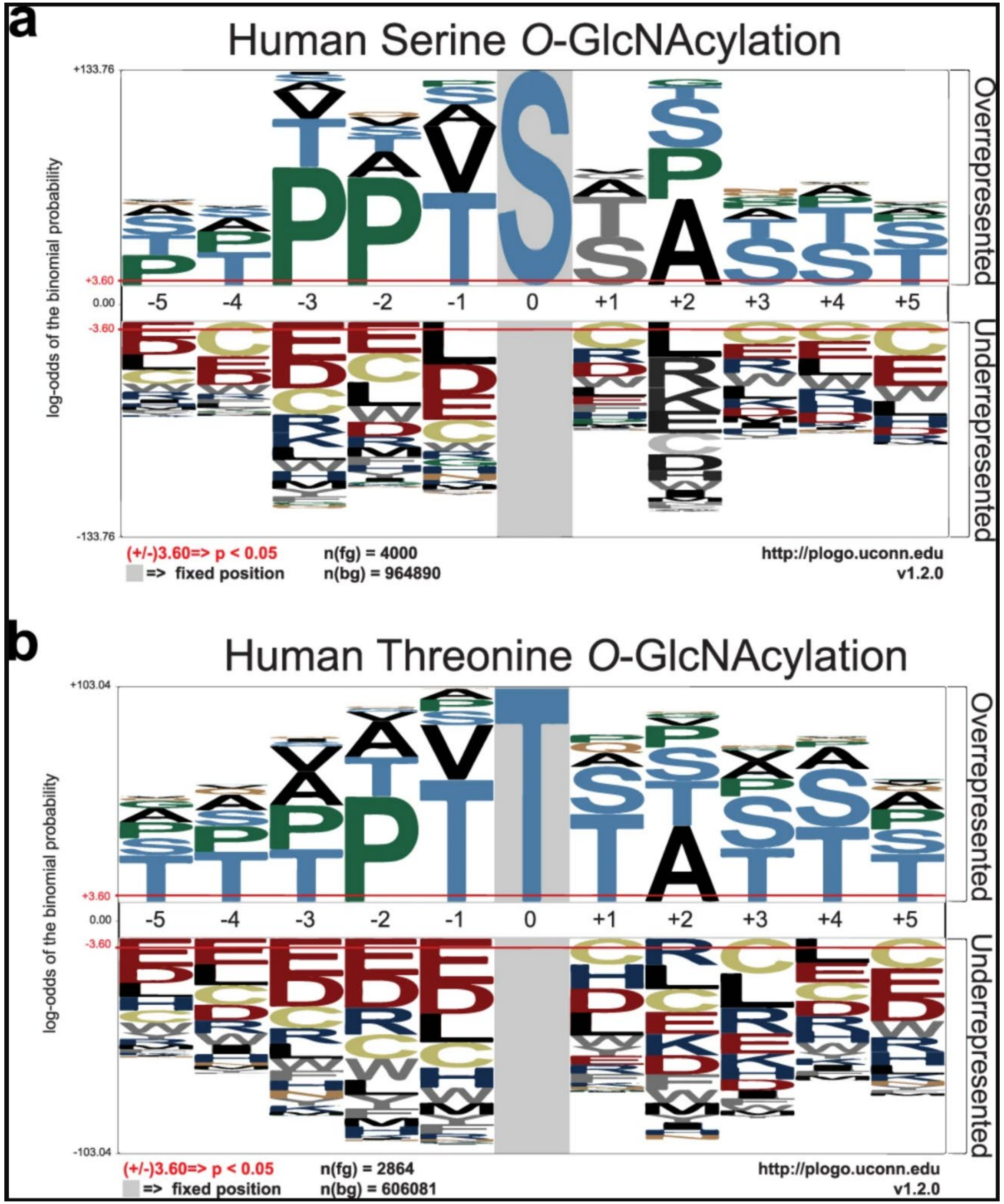


Wulff-Fuentes et al., 2021

Glycosylation Sites

N-linked glycosylation site:

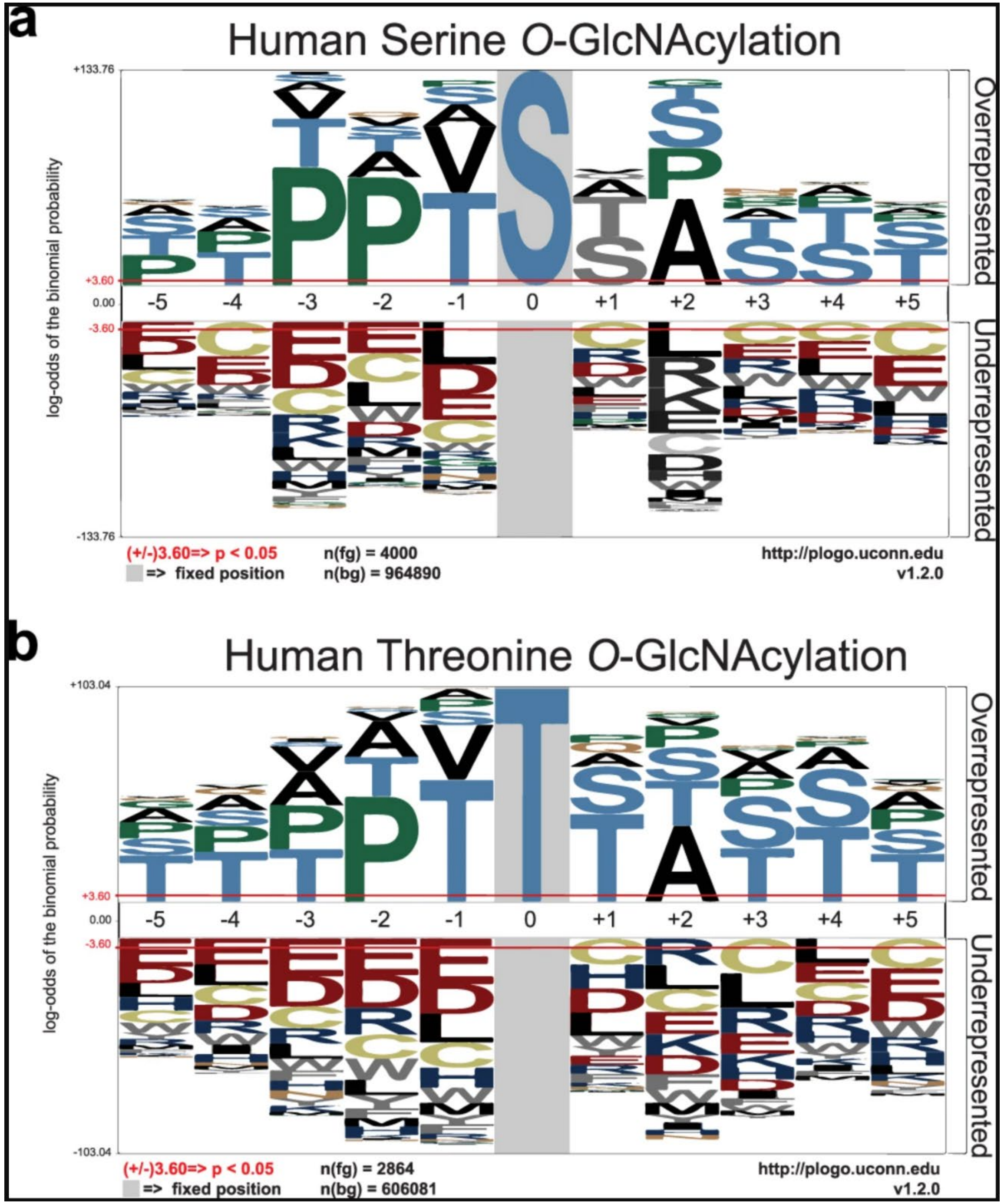
PNGase F recognition site:

NXS/T

X = anything but proline

O-linked glycosylation site:

Serine:
P-P-(V/T)-g(S)-(S/T)-A
Threonine:
(P/T)-P-(V/T)-g(T)-(S/T)-(A/T)



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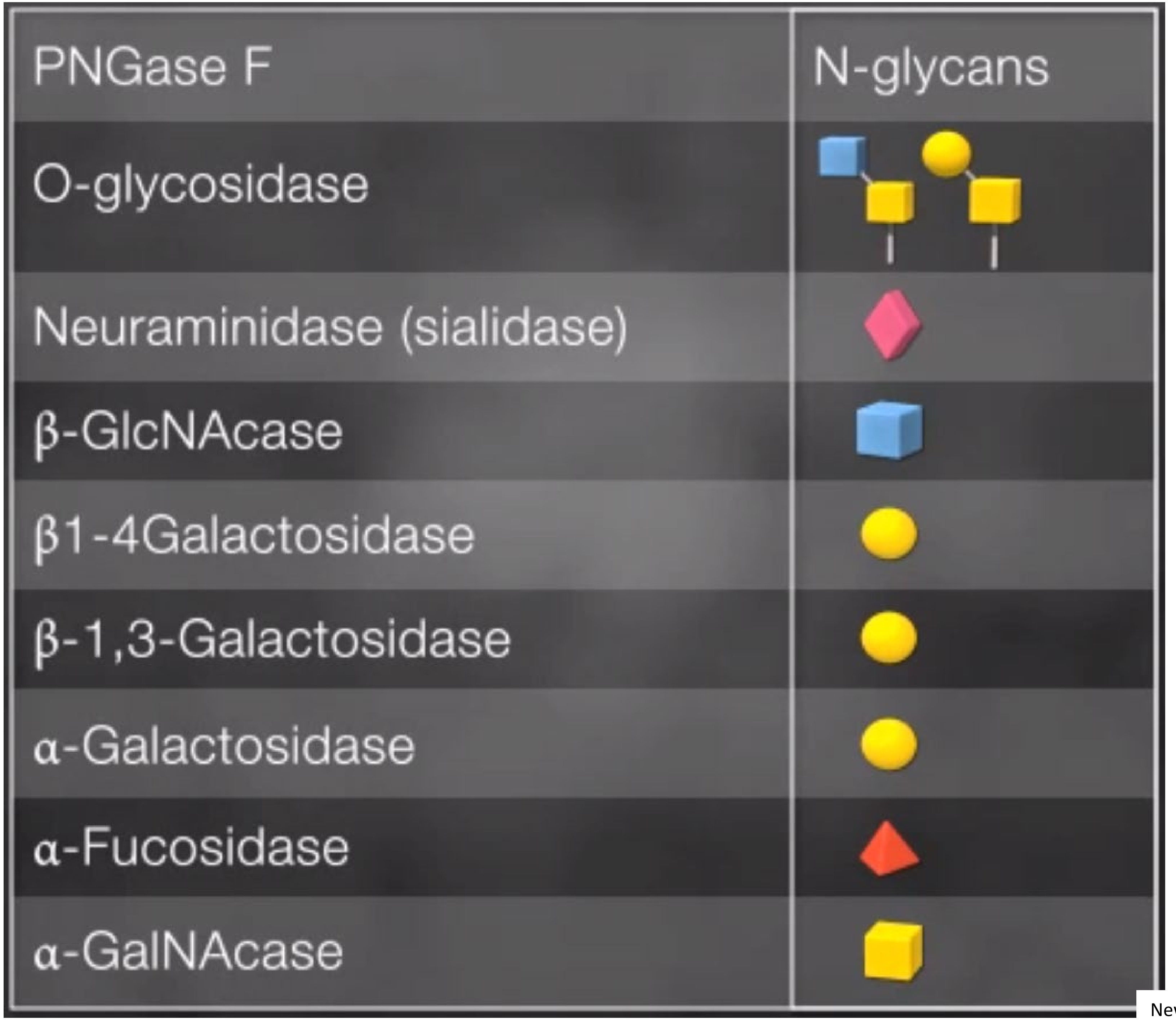
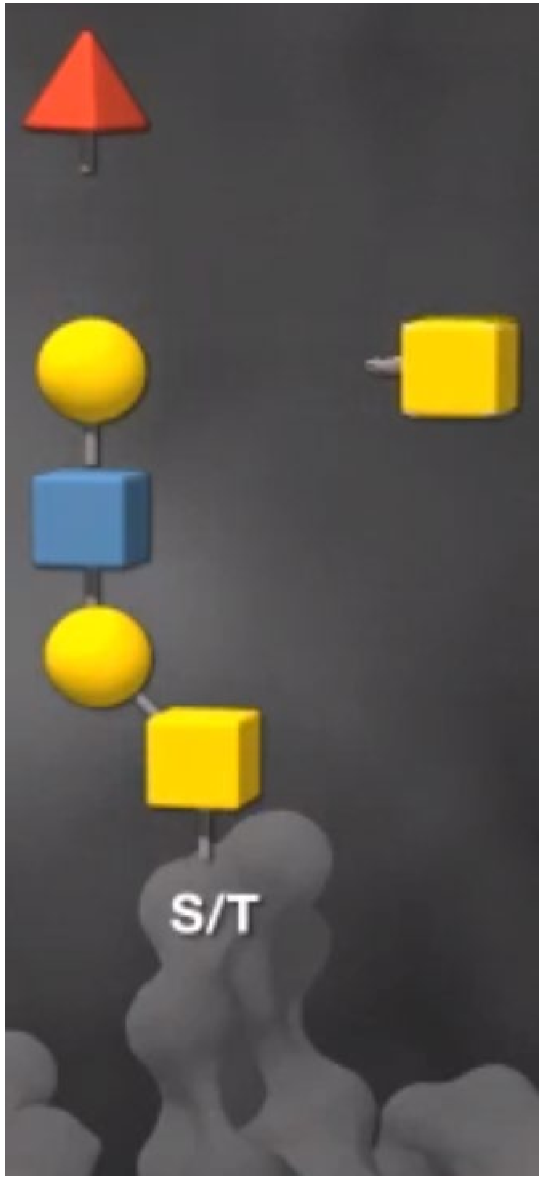


PNGase F	N-glycans
O-glycosidase	
Neuraminidase (sialidase)	
β-GlcNAcase	
β1-4Galactosidase	
β-1,3-Galactosidase	
α-Galactosidase	
α-Fucosidase	
α-GalNAcase	



New England Biolabs: https://www.neb.com/products/glycobiology/glycobiology

	PROTEIN DEGLYCOSYLATION MIX II (#P6044)	O-GLYCOSIDASE (#P0733 & #E0540)	PNGASE A (#P0707)	PNGASE F (#P0704 & #P0705)	REMOVE-IT PNGASE F (#P0706)	RECOMBINANT PNGASE F (#P0708 & #P0709)	RAPID PNGASE F (#P0710)	RAPID PNGASE F, (NON-REDUCING FORMAT) (#P0711)	ENDO H (#P0702) & ENDO HF (#P0703)	ENDO S (#P0741)	ENDO D (#P0742)	ENDO F2 (#P0772)	ENDO F3 (#P0771)
Deglycosylation of glycoproteins (N- and O-glycans)	●												
Removal of O-glycans	●	●											
Removal of N-glycans from glycoproteins	●		●	●	●	●	●	●	●	●	●	●	●
Removal of high mannose and hybrid N-glycans (leaving a GlcNAc attached to Asn)									●				
Optional removal of the enzyme from the reaction					●					●	●	●	●
Removal of paucimannose N-glycans (GlcNAc attached to Asn)						●			●		●		
Removal of N-glycans from IgGs (leaving a GlcNAc attached to Asn)									●	●			
Analysis of therapeutic glycoproteins, compliance with regulatory agencies						●	●	●					
High throughput N-glycan analysis of monoclonal antibodies, regulatory compliance							●	●					
Glycomics				●	●	●	●	●	●	●	●	●	●
Proteomics				● (only GF)	●	● (only GF)	●	●	●	●	●	●	●
Determine N-glycan sites									●	●	●	●	●
Removal of N-glycans from plant and insect			●										

New England Biolabs: https://www.neb.com/tools-and-resources/selection-charts/endoglycosidase-selection-chart

Signature of PNGase F Treatment

- PNGase F recognition site: NXS/T
  - X = anything but proline
- PNGase F deamidates Asparagine, resulting in an Aspartic Acid residue
  - N → D
  - $-NH_2 \rightarrow -OH = +0.984$ Da

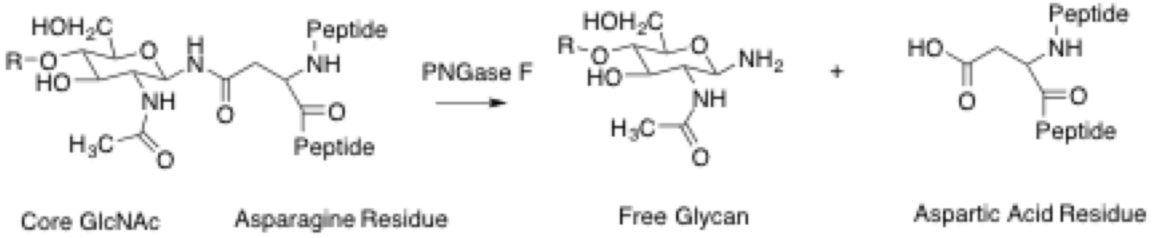


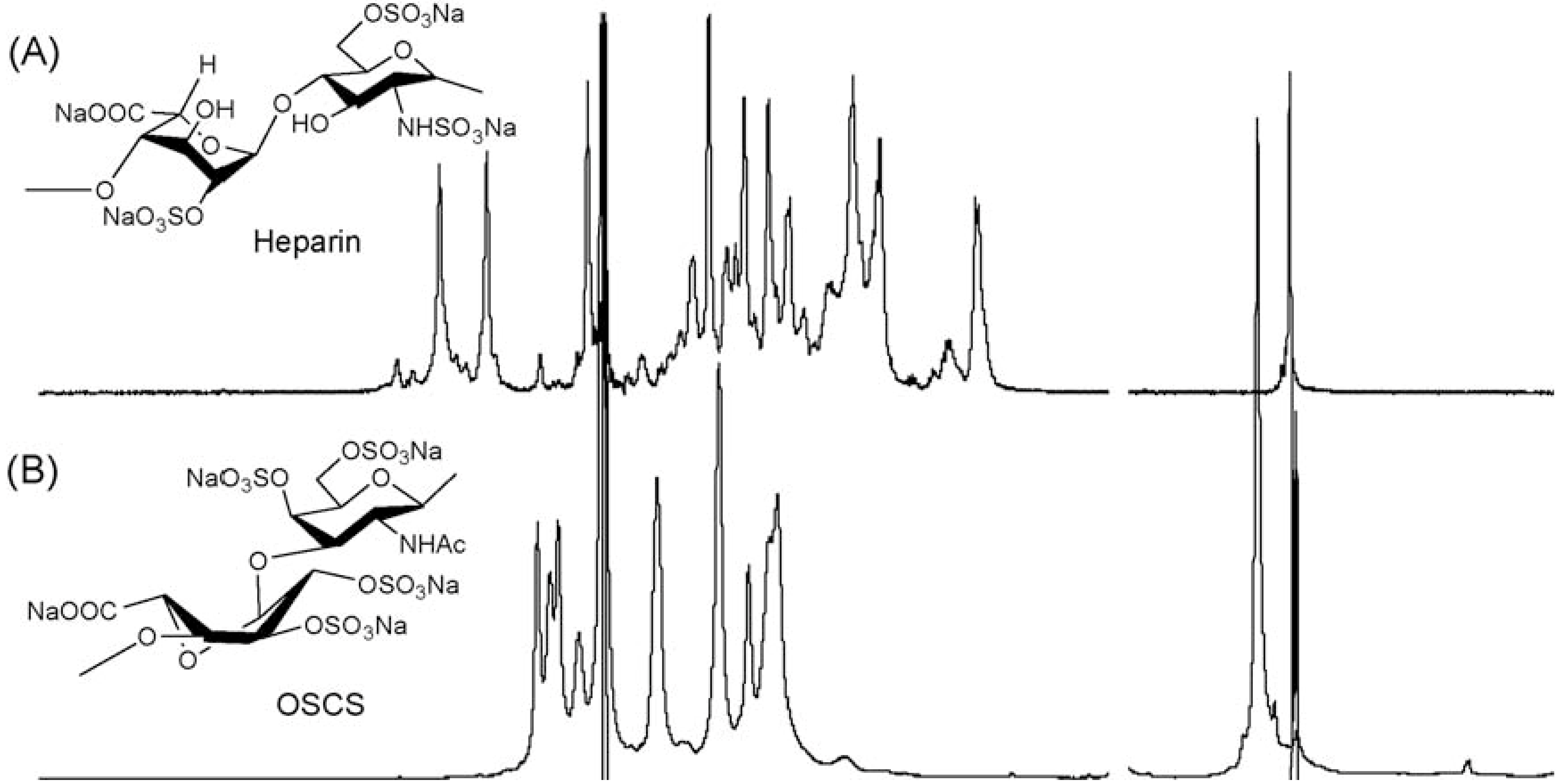
Image credit: Wikipedia https://en.wikipedia.org/wiki/PNGase_F

Glycoproteomics Exercise: Background

- Therapeutic glycoproteins are an important segment of biotherapeutics, biosimilars, and vaccines
- Enzymatic construction of glycans requires the full complement of human (or human-like) cellular glycosylation machinery to make proper glycoproteins
  - Strategies for commercial production of therapeutic glycoproteins proposed, but risky (e.g. Heparin)
  - Alternative strategy: obtain from donor population

Challenges to glycoprotein therapeutics example: Heparin

- Heparin
  - Glycosaminoglycan
  - Very safe
- Over-sulfated chondroitin sulfate
  - Contaminant
  - Unsafe



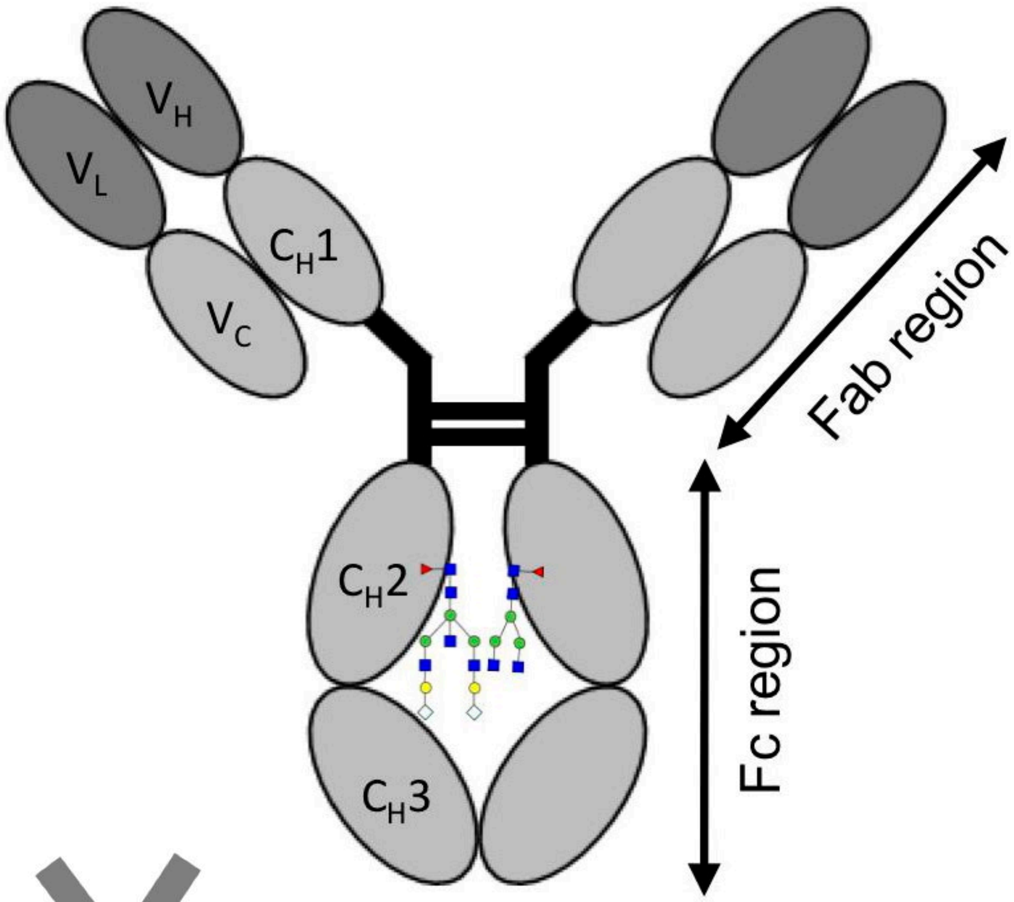
Li et al. Biochemical Pharmacology, 2009. DOI:10.1016/j.bcp.2009.04.012

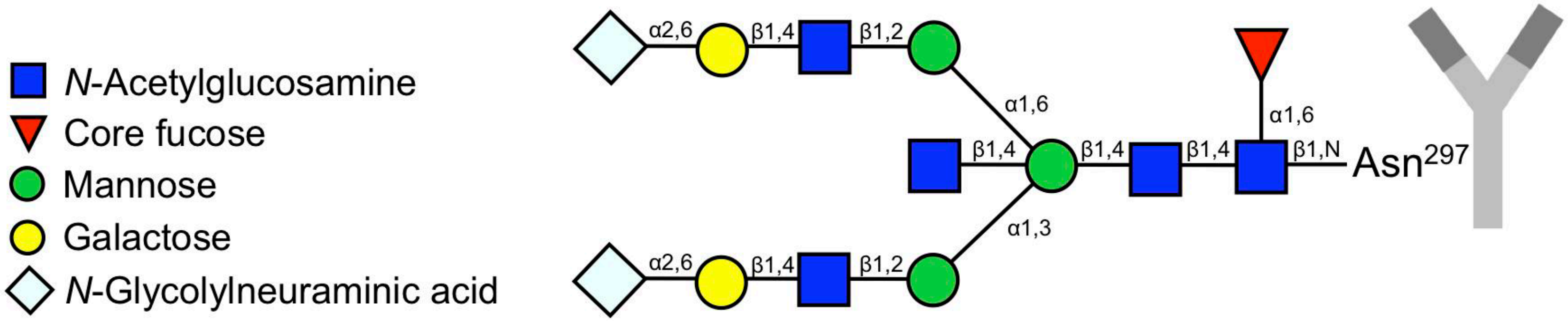
Motivation: Glycosylation analysis of human glycoprotein IgG1

- Intravenous Immunoglobulin G (IVIG) is the leading product obtained by fractionation of human plasma
  - Standard replacement therapy for primary and secondary immune deficiencies
- IVIG is extracted from the pooled plasma of thousands of human donors
  - Presumed to represent normal, healthy individuals
- Extracted IgG represents a cross-section of a healthy immunological response to cumulative exposure of donor population to environment
  - Specific composition of commercially available IVIG depends on donor pool and its specific environment

IgG1 Glycosylation

- Fc region of IgG1 has an N glycosylation site
- Variety of N glycans at this site are believed to play a crucial role in its effector functions
    - Includes anti inflammatory effects
    - Target for fractionation and isoform enrichment
    - Step 1: Look for N-glycosylation occupancy in the general public
        - Start with IgG1 (P01857)



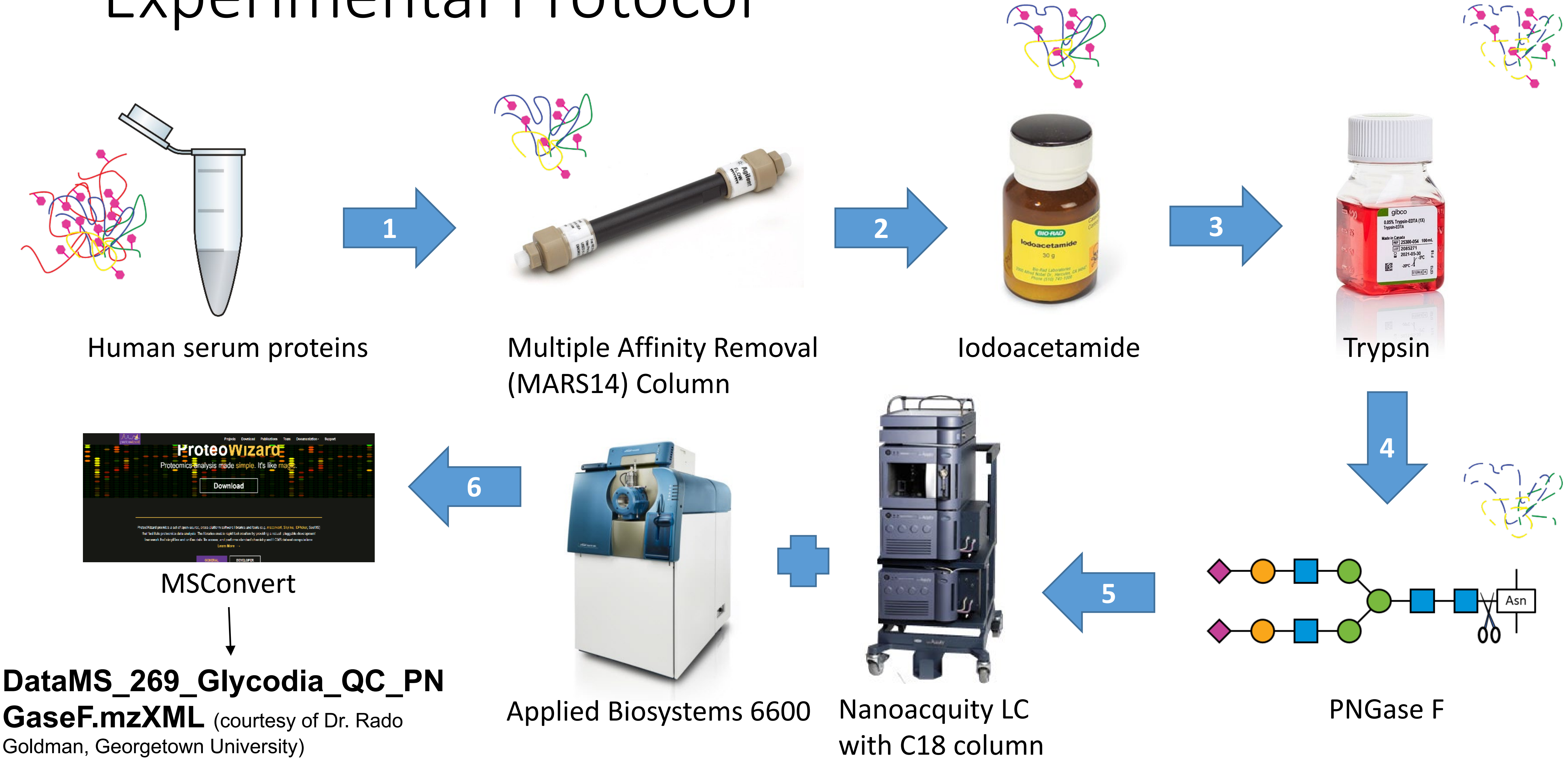


Zaytseva et al. Front. Cell Dev. Biol. 2020.

Biological Question

- In the general population, are the N-glycosylation sites in IgG1 (P01857) **fully occupied, partially occupied, or not occupied**?

Experimental Protocol



Tandem Mass Spec

- Ionize sample
- Mass spec 1: Quadrupole
  - Select specific range of peptides
- Collision chamber
  - Produce small fragments
- Mass spec 2: Time of Flight
  - Detect mass:charge ratios

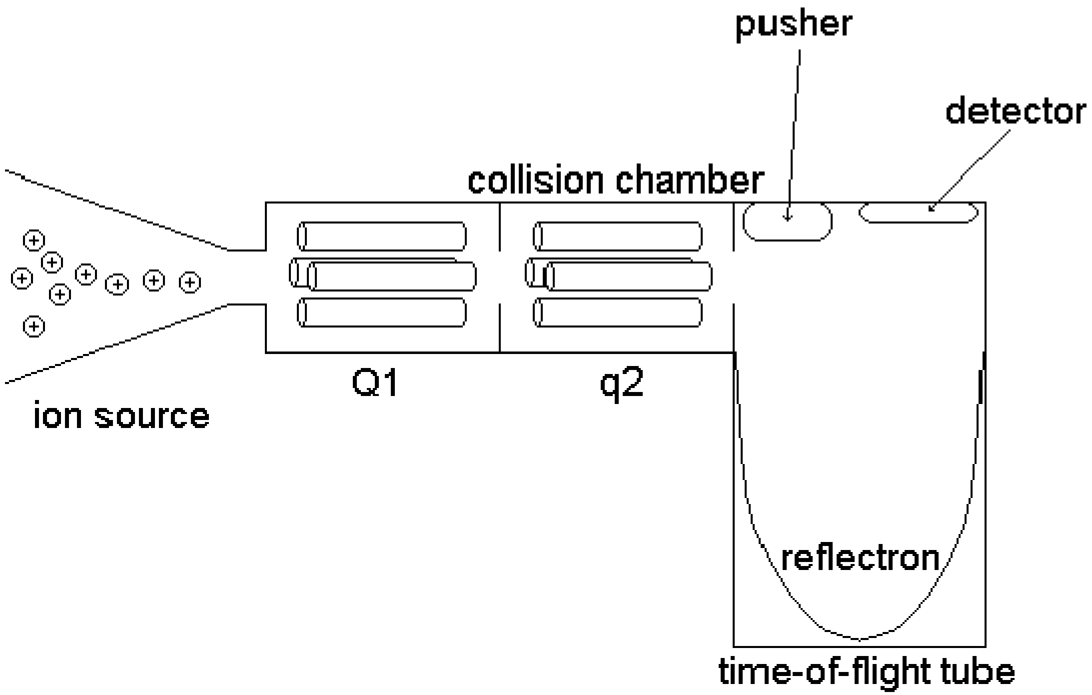
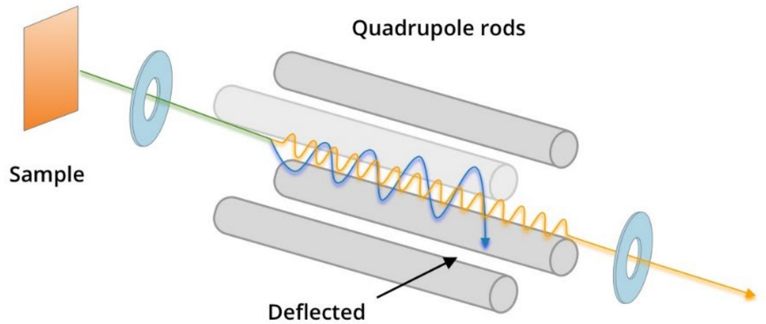


Figure: Yanofsky, C. A Bayesian approach to peptide identification by accurate mass and time tags in proteomics experiments. 2009. McGill University Press. Access via ResearchGate.

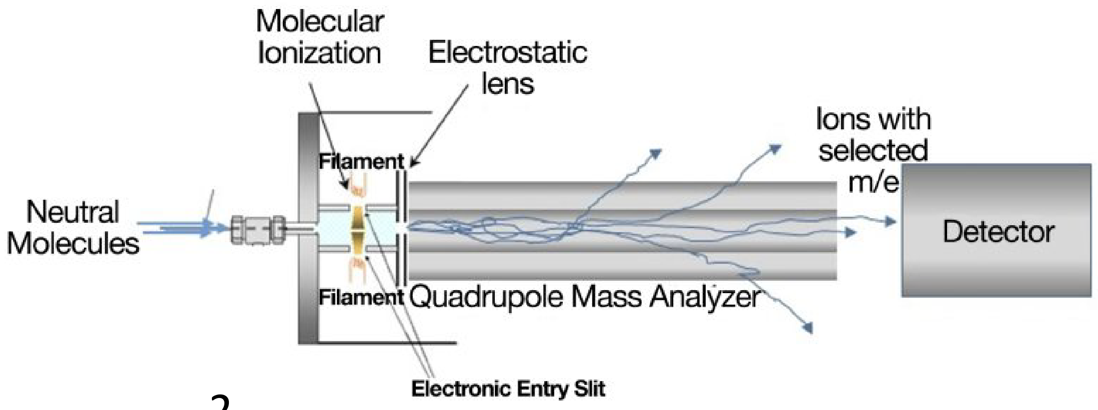
Figure 1: https://myscope.training/SIMS_Mass_analysers_in_SIMS
Figure 2: https://www.mks.com/n/mass-spectrometry

Quadrupoles Select a Specific m/z

- Create a fluctuating electric field
- The stability of an ion passing through an electric field is dependent on its mass:charge ratio
- Voltage defines the electric field
  - We control the voltage
- Therefore, we can control which mass:charge ratios are stable and make it through the field



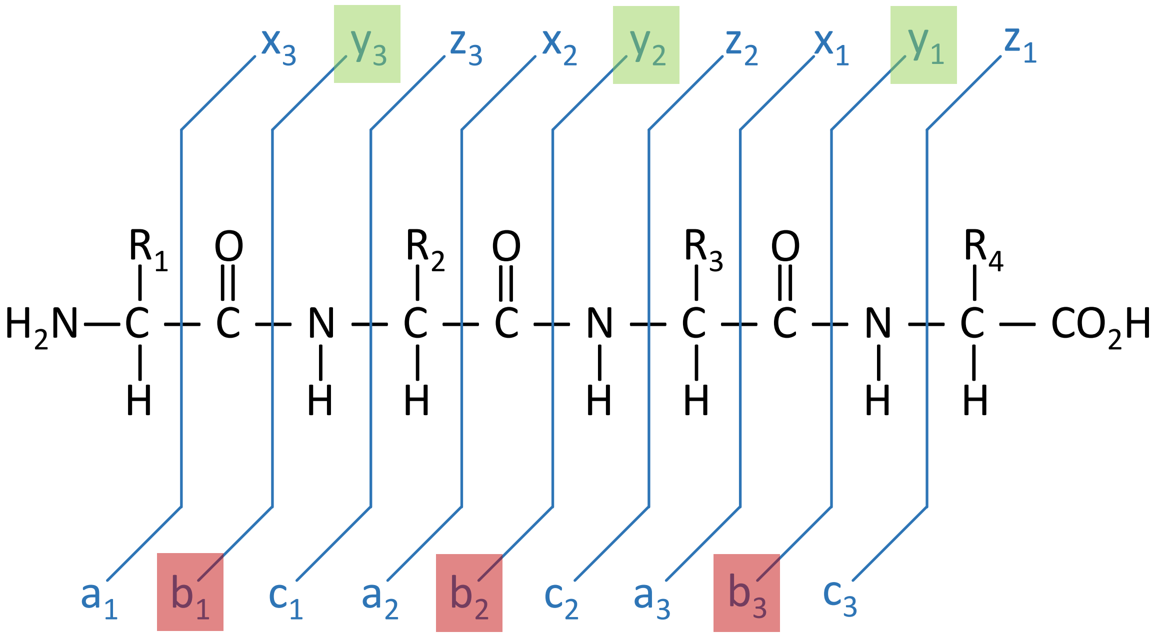
1



2

Spectra Generation

PTLN



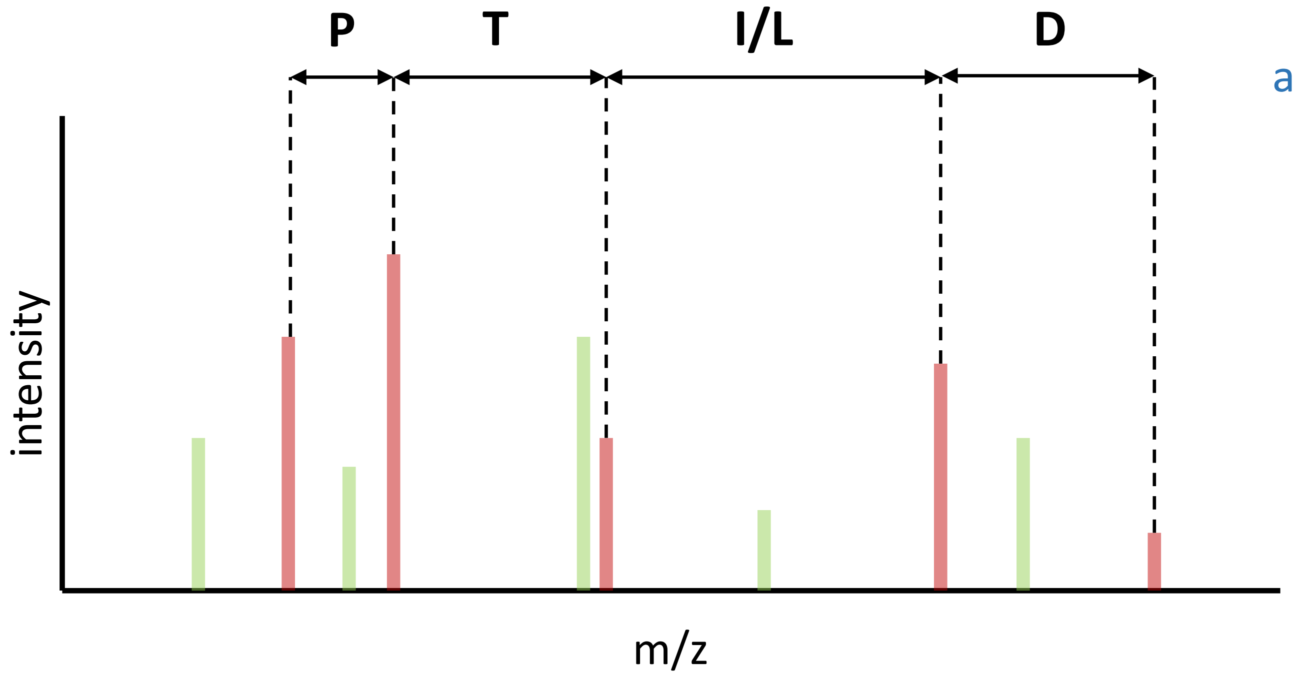
E.g.

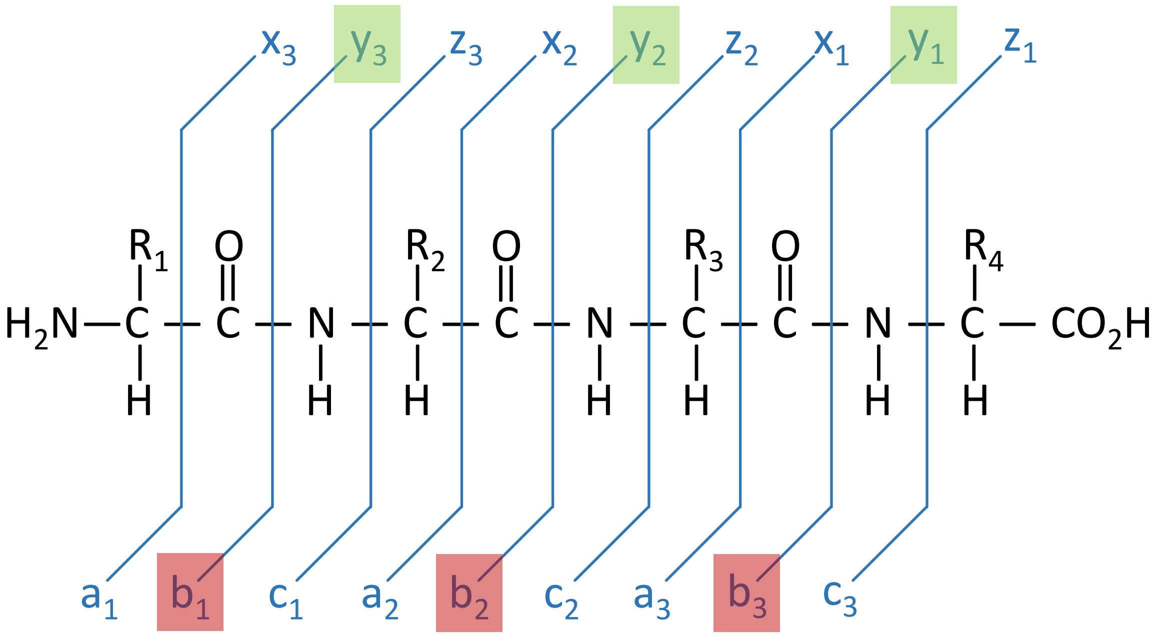
$^{+}$PT → b_2

TLN^{+} → y_3

After Doucette, A.

Spectra Generation





After Doucette, A.

Biological Question

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What do we expect to see?

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What we know:

1) We know that PNGaseF will convert an asparagine into an aspartic acid if there was a glycan at that position
2) We know that the difference between these two amino acid residues is 0.98 daltons
3) A mass spectrometer determines the mass of a molecule
   - (by measuring the mass-to-charge ratio (*m/z*) of its ion)

What do we expect to see?

- Peptides from proteins with N-glycans attached will be observed with a +0.98 Da mass-shift at Asn N-glycosites
  - Any *NXS/T* site will show a mass shift of +0.98 Da at the N if a glycan was present
  - If every sequence has this mass shift, there was full occupancy

  - **MS-GF+** is used to identify tryptic peptides with (and without) this Asn modification, providing evidence for N-glycosites with high, partial, or low occupancy
    - Be aware: MS-GF+ will not only look for the mass shift at the motif sequence, it will look for it everywhere!

What peptide(s) do we expect to see?

- Search protein sequence for motif
  - Fragment sequence by trypsinization (cut at lysine and arginine)
    - On carboxy side
  - Within those fragments, find NXS/T

Search for Glycosylated Sequence

- Can use an online motif or pattern recognition search
- Can write your own
- Can use a simple regex search:
- Go to https://regex101.com/
  - Input sequence from Uniprot `P01857`
    - Remove newline characters
  - Search for `N[^P][S|T]|K|R`

`N`	Literal \`N\` character
`[^P]`	Not the \`P\` character
`[S|T]`	One character, either S or T, nothing else
`K`	Literal \`K\` (lysine, a residue that trypsin cuts at)
`R`	Literal \`R\` (arginine, a residue that trypsin cuts at)

Advanced Search

```
(?:^|(?<=[KR])(?!P))(?:[^KR]|[KR](?=P))*N[^P][ST](?:[^KR]|[KR](?=P))*(?:(?=[KR](?!P))|$)
```

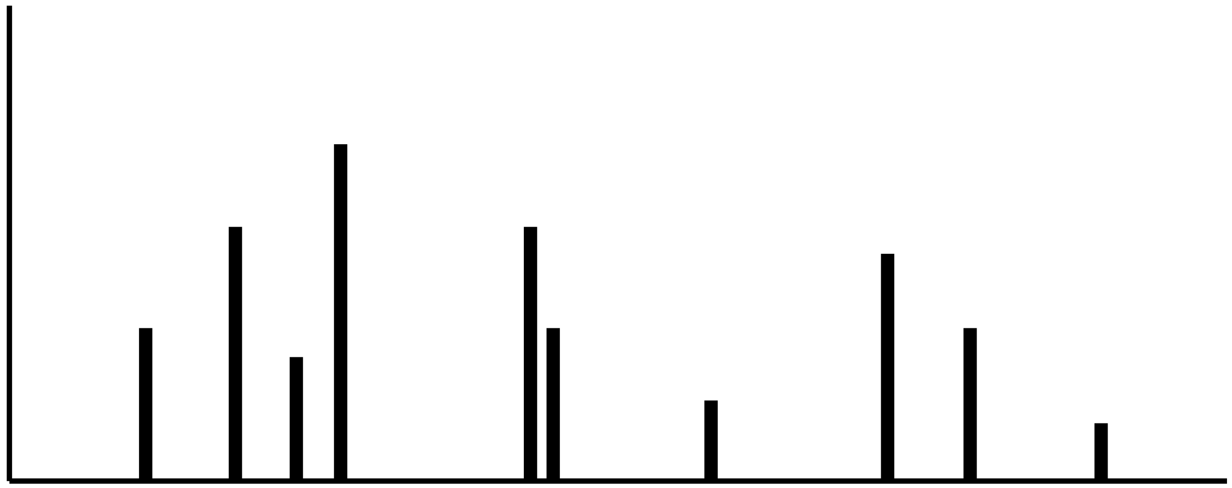
- Allows peptide to start at beginning or right after a cleavage site
- Enforces that cleavage sites are K/R not followed by P
- Finds N[^P][ST] inside
- Allows peptide to end at next cleavage site or end-of-sequence
- Works across newlines
- See regex101.com for detailed explanation of each clause

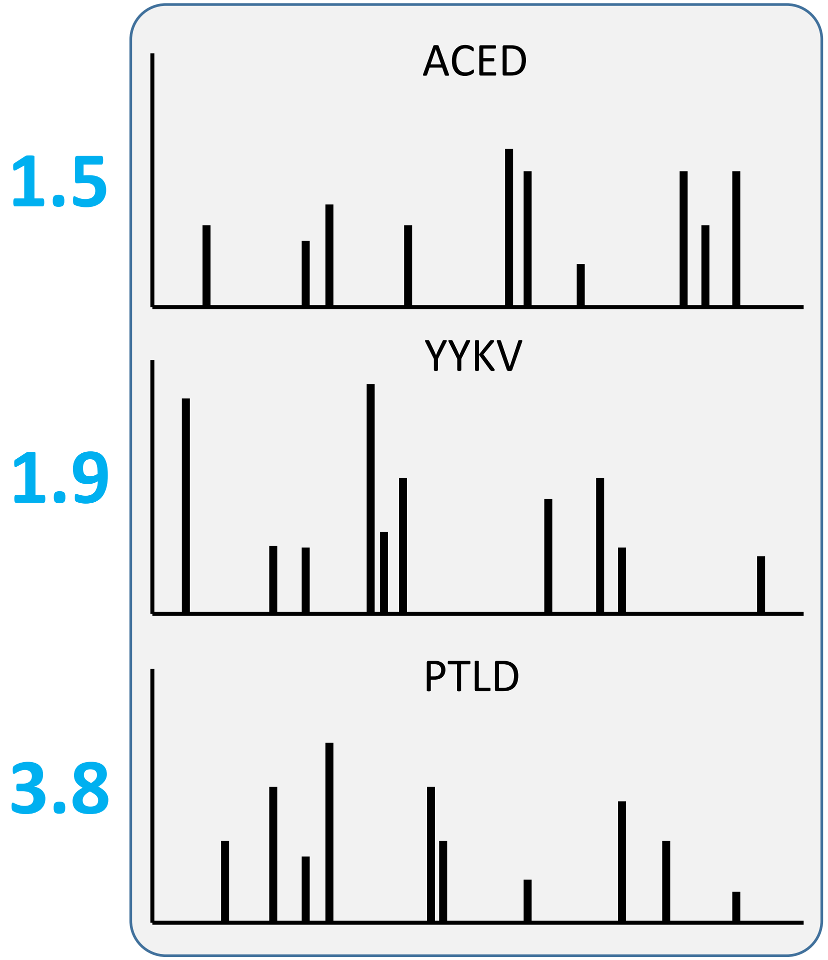
MS-GF+

- Database search tool
- Identify glycosylated (or otherwise modified) peptides in LC-MS/MS datasets
- Used to accurately identify peptides using an advanced search engine
- We will use to identify glycosylated peptides in our dataset:
  - **DataMS_269_Glycodia_QC_PNGaseF.mzXML**
- Using the UniProt reference:
  - **MARS14_proteins.fasta**

- MS-GF+: Universal Database Search Tool for Mass Spectrometry, Sangtae Kim, Pavel A. Pevzner, Nat Commun. 2014 Oct 31;5:5277. doi: 10.1038/ncomms6277.
- Spectral Probabilities and Generating Functions of Tandem Mass Spectra: A Strike against Decoy Databases, Sangtae Kim, Nitin Gupta and Pavel Pevzner, J Proteome Res. 2008 Aug;7(8):3354-63. doi: 10.1021/pr8001244.

Compare Unknown Spectra to Database of Theoretical Spectra





After [Doucette, A](#).

DataMS_269_Glycodia_QC_PNGaseF.mzXML

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      msLevel="2"
      peaksCount="39"
      polarity="+"
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      basePeakMz="4.760529676643e-05"
      basePeakIntensity="83.0"
      totIonCurrent="2566.0">
  <precursorMz precursorIntensity="0" activationMethod="HCD" windowWideness="1.0">400.672919771604</precursorMz>
  <peaks compressionType="zlib"
         compressedLen="627"
         precision="64"
         byteOrder="network"
         contentType="m/z-int">eJxziGV4MbF2w2X76dG…>
</scan>
```

Second MS

DataMS_269_Glycodia_QC_PNGaseF.mzXML


```
    <scan num="2889"
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Second MS

How many peaks are in this data?

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How many peaks are in this data?

How to decompress (MS data is huge)

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Byte64 encoded (to get text characters instead of binary values – XML can only represent text)

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Second MS

Data is interspersed values: m/z with intensity

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</scan>
```

Actual data

Data is interspersed values: m/z with intensity

Second MS

How many peaks are in this data?

How to decompress (MS data is huge)

Byte64 encoded (to get text characters instead of binary values – XML can only represent text)

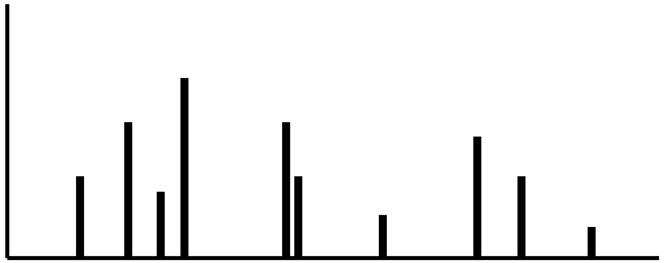
MS-GF Steps

- Base64 decode the string
- zlib decompress
- Read in pairs:
  - m/z
  - intensity

```
m/z₁, intensity₁,
m/z₂, intensity₂,
...
```

MS-GF Steps

- Base64 decode the string
- zlib decompress
- Read in pairs:
    - m/z
    - intensity



```
m/z₁, intensity₁,
m/z₂, intensity₂,
...
```

Scoring

- For this scan:
  - Precursor m/z = 400.67292
  - Charge = ???
- MS-GF+ computes candidate peptide neutral masses (within some tolerance)
- For each candidate:
  - Computes theoretical b/y ion masses
  - Compares to the 39 observed fragment m/z values
  - Scores the match probabilistically

Scoring

$$M = \left(\frac{m}{z} \cdot z\right) - z \cdot 1.007276$$

For z = 2:

$$M \approx (400.6729 \cdot 2) - 2 \cdot 1.007276$$
$$M \approx 801.3458 - 2.014552$$
$$\boxed{M \approx 799.3313 \text{ Da}}$$

For z = 3:

$$M \approx (400.6729 \cdot 3) - 3 \cdot 1.007276$$
$$M \approx 1202.0187 - 3.021828$$
$$\boxed{M \approx 1198.9969 \text{ Da}}$$

- For this scan:
  - Precursor m/z = 400.67292
  - Charge = ???
- MS-GF+ computes candidate peptide neutral masses (within some tolerance)
- For each candidate:
  - Computes theoretical b/y ion masses
  - Compares to the 39 observed fragment m/z values
  - Scores the match probabilistically

Calculate FDR

- Build a “decoy” database
  - Reverse all of the peptide sequences
- Scan decoy database for hits
  - Any hits are false positives
- Estimate error rate
  - Decoy hits/target hits
    - E.g. 10/990 ≈ 1%

```
MKWVTFISLLFLFSSAYSRGVFRRDTHKSEIAHRFKDLGE
```

```
EGLDKFRHAIESKHTDRRFGVRSYASSFLFLLSIFTVWKM
```

Homework:
(Due by February 23rd at 11:59PM Eastern)

- Is/are the N glycosite(s) in IgG2 (P01859) **fully occupied**, **partially occupied**, or **not occupied**?
  - Upload a Word doc that gives the answer and explains why
    - What is/are the peptide fragment(s) with N-glycosites?
    - For each glycosite, what is the occupancy in this dataset?
    - Remember to account for isotope errors and quality scores, if relevant
  - Use data from the tsv table to support your answer (as a figure or table)
  - Use the data set that was computed in class on Cerberus